

TraLO evidence and paper-claim boundary

Research status through 4 October 2026

Abstract

Completed studies support a bounded development-set finding for a snapshot side-copy intervention on knee radiographs. They do not establish that training-time dual-constraint TraLO beats a matched post-hoc clipper without secondary tradeoffs or on untouched data. This is a reproducibility and claim-readiness report, not held-out confirmation.

1 Question and intervention

The maintained question is whether training-time TraLO improves deployed constrained-class F1 (cc-F1) over a matched post-hoc clipper without being dominated on the rest of the profile. The positive knee study instead trains the base model once per seed, applies a targeted constraint step to side copies of saved epoch models, and averages their probabilities. Controls are an unstepped snapshot ensemble and a same-radius random-direction sham. All use the same capped-first allocator with 76 grade-3 slots. This is a side-copy intervention, not a training-time dual-constraint loss result.

2 Completed results

Knee backbone	Seeds	TraLO-PTO cc-F1	95% paired seed CI
ResNet18	72	+1.11 pp	[+0.78, +1.45] pp
RegNetY-400MF	72	+0.93 pp	[+0.57, +1.29] pp
MobileNetV3-Large	72	+1.18 pp	[+0.84, +1.51] pp
EfficientNet-B5	48	-0.87 pp	[-1.52, -0.22] pp

These prespecified E2 contrasts use a two-contrast Holm family within each backbone. ResNet18 and RegNetY also lost weighted-F1 against PTO: -0.25 $[-0.48, -0.03]$ and -0.52 $[-0.84, -0.20]$ percentage points. Thus their cc-F1 gains are not all-metric wins; B5 reverses the primary effect. All knee outcomes are on an inspected development pool. Seed intervals do not measure variation over unseen patients or sites. The 1,656-image Chen test remains unopened.

The fMoW cap-10 TraLO-minus-focal-clipper class-conditional F1 was -0.01393 with interval $[-0.01964, -0.00821]$ on viewed development countries. The reserved countries remain unopened.

For the calibrated four-seed MobileNetV3 cells, allocated TraLO-minus-PTO cc-F1 was:

- ISIC 2020 level 1: -0.0061 (CI -0.0707 to $+0.0586$).
- ISIC 2020 level 2: $+0.0365$ (CI -0.0496 to $+0.1225$).
- CelebA level 1: $+0.000045$ (CI -0.000140 to $+0.000230$).
- CelebA level 2: $+0.000079$ (CI -0.000951 to $+0.001110$).

Neither exploratory cell demonstrates a positive local dual-constraint effect.

3 Code and artifact validation

Release `0d90bc6bff90fdf7555f19c00061e0e5b0426147` passed the complete local suite (646 passed, one skipped, 42 subtests) and focused CPU tests on both DSI hosts with GPUs hidden. Git identity and committed SHA-256 bytes match the immutable release on both hosts. CPU replay of all four complete knee blocks reproduced every archived outcome line exactly; the four error logs were empty. A separate read-only implementation at release `5f74ce387a48cb22e69742c3d8db71a4ead95c62` recomputed capped-slot allocation, class metrics, paired intervals and Holm correction from the saved snapshot tensors. Its expanded local suite passed (649 passed, one skipped); focused CPU tests passed on `dsisco01`. Direct `dsisco02` access was unavailable for this second release. An off-host comparison checked 96 summary fields and 792 per-seed cc-F1 values against archived reports within published rounding, including the B5 reversal and weighted-F1 harms. This is arithmetic validation on the same development pool, not new evaluation.

The old snapshot pilot gate called a helper that internally read development labels and computed metrics, although it printed no pilot score. We found no evidence of pilot-quality-based tuning, but the old gate was not technically label-blind. The new release separates the integrity gate from label-based scoring, detects a corrupt side snapshot, and checks every fixed seed before opening labels. Historical runs and results remain unchanged.

4 Open gates

A stronger paper claim needs a frozen training-time method, matched controls, allocator, caps, primary contrast, secondary tradeoff rule, and analysis plan before sealed labels are opened. Opening the Chen test or reserved countries requires a user decision under project rules. The proposed HAM10000 melanoma-versus-rest cell changes the scientific question and is not approved; patient IDs are unavailable. ISIC 2024 still lacks a frozen patient-disjoint split and several data and training gates. The verified nonoverlapping compute spend is at least 72.175499 GPU-hours of a finite 96-hour ceiling; 23.824501 hours is only an upper bound on unspent budget, not a certified balance.

5 Evidence locations

The detailed audit and numerical sources are `experiments/paper_claim_readiness_20261004.md`, `experiments/tralo_paper_results_draft_20260930.md`, the preregistered reports under `experiments/`, and scorer outputs under `analysis/`. The older incomplete overnight report entry point is preserved at `report/archive/overnight_20260927_main.tex`.